1. Who developed the first electrochemical battery (the Voltaic Pile) in 1800 producing continuous direct current?

(A) Alessandro Volta
(B) Luigi Galvani
(C) Michael Faraday
(D) Andre-Marie Ampere

2. Which fundamental law discovery by Michael Faraday in 1831 forms the basis of all electromagnetic generators and transformers?

(A) Faraday's Law of Electromagnetic Induction
(B) Ohm's Law
(C) Coulomb's Law
(D) Ampere's Circuital Law

3. The historic late 19th-century commercial competition between Edison's DC system and Westinghouse-Tesla's AC system is known as:

(A) The War of the Currents
(B) The Power Struggle
(C) The Grid Revolution
(D) The Electrification Race

4. What crucial technical invention allowed AC power to be transmitted over long distances far more efficiently than DC in the 1890s?

(A) The AC Transformer
(B) The Mercury Arc Rectifier
(C) The Vacuum Tube Diode
(D) The Lead-Acid Battery

5. In 1895, the first large-scale multi-phase alternating current hydroelectric generating station in North America was built at:

(A) Niagara Falls
(B) Hoover Dam
(C) Grand Coulee Dam
(D) Tennessee Valley

6. Who patented the polyphase AC induction motor and rotating magnetic field principles in 1888?

(A) Nikola Tesla
(B) Thomas Edison
(C) George Westinghouse
(D) Charles Proteus Steinmetz

7. What is the standard commercial AC power transmission frequency in India, Europe, and most of Asia?

(A) 50 Hz
(B) 60 Hz
(C) 25 Hz
(D) 100 Hz

8. What is the standard commercial power frequency used across the United States and Canada?

(A) 60 Hz
(B) 50 Hz
(C) 40 Hz
(D) 120 Hz

9. What primary advantage did high-voltage AC transmission provide over low-voltage DC in the early grid development?

(A) Lower $I^2 R$ transmission line losses over long geographical distances
(B) Zero reactive power
(C) Simpler underground cabling
(D) No need for circuit breakers

10. Which scientist first mathematically formulated the complete set of classical electromagnetic field equations?

(A) James Clerk Maxwell
(B) Heinrich Hertz
(C) Oliver Heaviside
(D) William Thomson

11. Which electrical engineer developed the complex number (phasor) method for analyzing AC circuits at General Electric?

(A) Charles Proteus Steinmetz
(B) Nikola Tesla
(C) Thomas Edison
(D) Frank Sprague

12. What electrical machine converts mechanical energy from a rotating turbine into alternating electrical power?

(A) Synchronous Alternator (Generator)
(B) Induction Motor
(C) Rotary Converter
(D) Autotransformer

13. What is the primary role of a step-up transformer located at an electrical power generating station switchyard?

(A) To increase voltage and proportionally decrease current for low-loss transmission
(B) To convert AC to DC
(C) To regulate grid frequency
(D) To provide reactive power compensation

14. What is the primary role of a step-down distribution transformer located near residential consumers?

(A) To reduce transmission voltage to safe utilization levels (e.g. 415V/230V)
(B) To increase power factor
(C) To store magnetic energy
(D) To filter harmonics

15. What constitutes a 3-phase, 4-wire electrical distribution system?

(A) Three phase conductors (R, Y, B) and one neutral conductor (N)
(B) Three hot lines and one ground shield
(C) Two phases and two neutrals
(D) Four independent DC lines

16. What is the standard phase displacement between the three voltages in a balanced 3-phase AC power system?

(A) 120 degrees electrical
(B) 90 degrees
(C) 180 degrees
(D) 60 degrees

17. What is the mathematical relationship between line voltage V_L and phase voltage V_{ph} in a balanced star (Y) connected system?

(A) $V_L = \sqrt{3} V_{ph}$
(B) $V_L = V_{ph}$
(C) $V_L = 3 V_{ph}$
(D) $V_L = V_{ph} / \sqrt{3}$

18. What is the mathematical relationship between line current I_L and phase current I_{ph} in a balanced star connected load?

(A) $I_L = I_{ph}$
(B) $I_L = \sqrt{3} I_{ph}$
(C) $I_L = I_{ph} / \sqrt{3}$
(D) $I_L = 3 I_{ph}$

19. In a balanced delta (Δ) connected load, what is the relationship between line voltage and phase voltage?

(A) $V_L = V_{ph}$
(B) $V_L = \sqrt{3} V_{ph}$
(C) $V_L = 3 V_{ph}$
(D) $V_L = V_{ph} / 2$

20. What is the primary international professional organization dedicated to electrical and electronics engineering?

(A) IEEE (Institute of Electrical and Electronics Engineers)
(B) ASME
(C) AIChE
(D) SAE

21. What does NBA stand for in Indian technical education quality accreditation?

(A) National Board of Accreditation
(B) National Bureau of Academics
(C) National Baseline Assessment
(D) New Board of Accreditation

22. What international agreement enables mutual recognition of accredited engineering degrees among member nations (including India)?

(A) The Washington Accord
(B) The Geneva Convention
(C) The Paris Treaty
(D) The Bologna Declaration

23. In outcome-based engineering education (OBE), what does 'PO' stand for?

(A) Program Outcomes
(B) Performance Objectives
(C) Practical Outputs
(D) Professional Overviews

24. What does 'CO' represent in an academic engineering course syllabus?

(A) Course Outcomes
(B) Class Objectives
(C) Curriculum Overview
(D) Core Operations

25. Which educational framework classifies cognitive learning into levels from Remember and Understand to Analyze and Create?

(A) Bloom's Revised Taxonomy
(B) Maslow's Hierarchy
(C) Kolb's Experiential Cycle
(D) Gagne's Nine Events

26. Which core pillar of Electrical Engineering focuses on rotating machinery, magnetic circuits, and electromechanical energy conversion?

(A) Electrical Machines
(B) Digital Signal Processing
(C) Very Large Scale Integration
(D) Antenna Design

27. Which core branch of EE deals with transmission lines, load flow, fault analysis, and grid stability?

(A) Power Systems
(B) Computer Architecture
(C) Software Engineering
(D) Solid State Physics

28. Which core EE subject focuses on solid-state power switches (IGBTs, MOSFETs, Thyristors) and energy conversion circuits?

(A) Power Electronics
(B) Analog Communications
(C) Microstrip Lines
(D) Database Management

29. Which branch of EE deals with feedback systems, stability criteria (Bode, Nyquist), and automated regulation of dynamic processes?

(A) Control Systems
(B) Thermodynamics
(C) VLSI Testing
(D) Compiler Design

30. What is the primary function of an electrical protection relay in a substation?

(A) To sense abnormal fault conditions and signal the circuit breaker to trip
(B) To generate electrical power
(C) To store electricity
(D) To measure atmospheric temperature

31. What mechanical device physically interrupts high fault currents in an electrical power network?

(A) High-Voltage Circuit Breaker
(B) Isolator switch without arc chute
(C) Current Transformer
(D) Voltmeter

32. What medium is commonly used inside modern high-voltage circuit breakers to extinguish electrical arcs?

(A) Sulfur Hexafluoride (SF_6) gas or Vacuum
(B) Carbon Dioxide
(C) Compressed Oxygen
(D) Mineral Water

33. What electrical system transmits bulk electrical energy over thousands of kilometers using direct current?

(A) High-Voltage Direct Current (HVDC)
(B) EHV-AC
(C) Medium Voltage AC
(D) Single Phase DC

34. What is a major advantage of HVDC over HVAC for long-distance subsea or underground power transmission?

(A) No charging current / capacitive reactive power loss limitations
(B) Requires no converters
(C) Uses cheaper copper coils
(D) Operates without transformers

35. What is the primary greenhouse gas whose reduction is driving global energy decarbonization in power engineering?

(A) Carbon Dioxide (CO_2)
(B) Nitrogen (N_2)
(C) Argon
(D) Helium

36. Which renewable energy source converts incident sunlight directly into electrical DC energy using semiconductor p-n junctions?

(A) Solar Photovoltaic (PV)
(B) Geothermal
(C) Tidal Barrage
(D) Nuclear Fission

37. Which renewable technology harnesses aerodynamic lift on rotating blades to drive an electrical generator?

(A) Wind Turbines
(B) Solar Thermal Stirling
(C) Ocean Wave Oscillating Column
(D) Hydroelectric Pump

38. What term describes small-scale, localized power generation situated close to the electrical load (e.g. rooftop solar)?

(A) Distributed Generation (DG)
(B) Bulk Central Generation
(C) Base Load Thermal
(D) Pumped Hydro Storage

39. What is a self-sufficient energy system that serves a discrete footprint (like a university campus) and can operate in islanded mode?

(A) Microgrid
(B) Supergrid
(C) National Load Dispatch Center
(D) Switching Yard

40. What device converts variable DC output from solar panels into grid-synchronized 50/60 Hz AC power?

(A) Grid-Tied Solar Inverter
(B) Rotary Phase Converter
(C) DC Buck Chopper
(D) Autotransformer

41. Why does AC power transmission experience skin effect in conductors, and what is its consequence?

(A) AC current concentrates near conductor surface, increasing effective AC resistance above DC resistance
(B) AC current concentrates in the center
(C) Decreases power loss
(D) Eliminates inductance

42. What conductor type composed of aluminum strands surrounding a central steel core is standard in overhead transmission lines?

(A) ACSR (Aluminum Conductor Steel Reinforced)
(B) Pure Copper Bar
(C) Solid Lead Wire
(D) Tungsten Alloy

43. In a synchronized AC power grid, what parameter acts as an instantaneous indicator of the balance between total generation and total load?

- **(A)** System Frequency
- **(B)** Line Voltage at the generator bus
- **(C)** Ambient temperature
- **(D)** Earth resistance

44. What happens to the grid frequency if total consumer load suddenly exceeds total generator mechanical power input?

- **(A)** Grid frequency drops below nominal (e.g. < 50 Hz)
- **(B)** Grid frequency rises rapidly
- **(C)** Voltage becomes DC
- **(D)** Power factor becomes zero

45. What control mechanism automatically adjusts turbine governor valves to restore frequency within seconds of a load change?

- **(A)** Primary Frequency Control (Droop Governor Control)
- **(B)** Tertiary Reserve Dispatch
- **(C)** Manual Operator Switching
- **(D)** Capacitor Bank Disconnect

46. What is the primary purpose of bundled conductors in Extra High Voltage (EHV) transmission lines (e.g. 400 kV and 765 kV)?

- **(A)** To increase effective conductor diameter, reducing surface electric field and corona discharge losses
- **(B)** To increase line resistance
- **(C)** To reduce mechanical weight
- **(D)** To insulate conductors

47. What luminous violet glow phenomenon occurs on high-voltage conductors when air ionization threshold (~30 kV/cm) is exceeded?

- **(A)** Corona Discharge
- **(B)** Glowworm effect
- **(C)** Dielectric breakdown
- **(D)** Arc flash blast

48. What does the Ferranti Effect describe in long, unloaded or lightly loaded high-voltage transmission lines?

- **(A)** Receiving-end voltage rises higher than sending-end voltage due to line charging capacitance
- **(B)** Receiving-end voltage drops to zero
- **(C)** Power factor becomes unity
- **(D)** Current lags by 90 degrees

49. Which device is installed at substation buses to counteract Ferranti effect and absorb excess capacitive reactive power?

- **(A)** Shunt Reactor
- **(B)** Series Capacitor
- **(C)** Synchronous Booster
- **(D)** Current Limiting Resistor

50. What device is placed in series with long transmission lines to reduce effective inductive reactance and increase power transfer capability?

- **(A)** Series Capacitor
- **(B)** Shunt Reactor
- **(C)** Surge Arrester
- **(D)** Lightning Rod

51. According to the power-angle equation, what is the theoretical maximum active power P_{max} transmissible over a lossless line of reactance X?

- **(A)** $P_{max} = \frac{V_S V_R}{X}$ at power angle $\delta = 90^\circ$
- **(B)** $P_{max} = \frac{V_S V_R}{R}$
- **(C)** $P_{max} = V_S I_R$
- **(D)** $P_{max} = \frac{V_S^2}{2X}$

52. In power engineering, what does the term 'Base Load' plant refer to?

- **(A)** Power stations that run continuously at near-constant full output to satisfy minimum baseline demand (e.g. nuclear, supercritical thermal)
- **(B)** Peaking gas turbines
- **(C)** Rooftop solar
- **(D)** Emergency diesel sets

53. Which power plant type possesses the fastest startup and ramp rate, making it ideal for peak shaving and grid frequency regulation?

- **(A)** Open-Cycle Gas Turbine or Pumped Hydro Storage
- **(B)** Nuclear Power Station
- **(C)** Supercritical Coal Thermal
- **(D)** Lignite Fired Plant

54. What is the function of an Automatic Generation Control (AGC) system operating in secondary frequency control?

- **(A)** Centrally adjusts generator setpoints via Area Control Error (ACE) to restore nominal frequency and schedule tie-line exchanges
- **(B)** Trips circuit breakers
- **(C)** Manually cools transformers
- **(D)** Computes electricity billing

55. What instrument transformer steps down high transmission line currents (e.g. 2000 A) to standardized 1 A or 5 A for meters and relays?

- **(A)** Current Transformer (CT)
- **(B)** Potential Transformer (PT)
- **(C)** Autotransformer
- **(D)** Distribution Transformer

56. Why must the secondary winding of an energized Current Transformer NEVER be left open-circuited?

- **(A)** Infinite/extreme secondary voltage develops across the open terminals, causing insulation explosion and lethal shock hazard
- **(B)** It blows the primary fuse
- **(C)** It causes reverse power flow
- **(D)** It draws zero magnetizing current

57. What instrument transformer steps down high voltages (e.g. 220 kV) to 110 V for metering and distance protection?

(A) Potential Transformer (PT) / Capacitor Voltage Transformer (CVT)
(B) Current Transformer
(C) Grounding Transformer
(D) Booster Transformer

58. What does SCADA stand for in electrical grid automation and substation management?

(A) Supervisory Control and Data Acquisition
(B) System Control and Digital Architecture
(C) Substation Computer and Digital Automation
(D) Synchronous Control and Dispatch Algorithm

59. What communication protocol is globally standardized under IEC for Substation Automation Systems (SAS)?

(A) IEC 61850
(B) IEEE 802.11
(C) Modbus RTU only
(D) HTTP/2

60. In IEC 61850 substation architecture, what are GOOSE messages used for?

(A) Generic Object Oriented Substation Events for ultra-fast peer-to-peer relay trip signals (< 4 ms)
(B) Monthly billing logs
(C) Operator voice chat
(D) Firmware flashing

61. What does Phasor Measurement Unit (PMU) technology provide for Wide Area Monitoring Systems (WAMS)?

(A) GPS-synchronized time-stamped voltage and current phasors sampled at 50-60 frames per second
(B) Monthly energy consumption
(C) Mechanical turbine vibration logs
(D) Oil temperature graphs

62. In ABET/NBA accreditation, what is the primary purpose of Continuous Quality Improvement (CQI)?

(A) Systematic curriculum revision based on closing-the-loop assessment of student learning outcomes
(B) Reducing faculty salaries
(C) Increasing tuition fees
(D) Purchasing identical textbooks

63. In modern engineering pedagogy, what does 'Active Learning' emphasize over traditional passive lecturing?

(A) Student engagement via problem-solving, collaborative peer discussions, lab simulations, and inquiry-based design
(B) Silent note-taking
(C) Memorizing textbook tables
(D) Multiple choice drills only

64. What is a 'Capstone Design Project' in undergraduate Electrical Engineering curricula?

(A) A comprehensive multi-term design project synthesizing engineering sciences, economic feasibility, safety, and societal impact
(B) A 10-minute viva
(C) A written history essay
(D) A textbook reading test

65. What is the primary function of a power system State Estimator at a National Load Dispatch Center?

(A) Filters noisy telemetry measurements to compute the most statistically probable system state (V, θ at all buses)
(B) Estimates fuel prices
(C) Predicts weather 6 months ahead
(D) Controls generator speed directly

66. What is N-1 contingency criterion in power system transmission reliability planning?

(A) The system must remain fully stable and within thermal/voltage limits following the loss of any single critical component (line, transformer, or generator)
(B) Loss of all generators
(C) Operating without protective relays
(D) Shutting down the grid during rain

67. What causes transient overvoltages during lightning strikes on overhead transmission lines, and how are equipment protected?

(A) Steep atmospheric surge impulses propagating along the line; diverted to ground via Surge Arresters (Metal Oxide Varistors - MOVs)
(B) Low power factor; protected by capacitors
(C) Direct current build-up; protected by fuses
(D) Harmonic resonance; protected by filters

68. What does the IEEE Std 1547 define regarding distributed energy resources (DERs)?

(A) Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces
(B) Standard for household plugs
(C) Standard for electric motors
(D) Standard for incandescent bulbs

69. What is 'Duck Curve' in California and modern grids with massive solar PV penetration?

(A) A net load curve showing deep midday dip due to solar generation followed by steep evening ramp as solar drops and demand peaks
(B) A plot of transformer efficiency
(C) A water turbine blade curve
(D) A battery life cycle chart

70. What grid-supportive capability requires solar and wind inverters to remain connected during brief grid voltage dips?

(A) Low-Voltage Ride-Through (LVRT)
(B) Anti-Islanding Trip
(C) Emergency Load Shedding
(D) Black Start Shutdown

71. What safety protection feature must automatically disconnect a grid-tied inverter if the utility grid loses power, preventing energized lines?

(A) Anti-Islanding Protection
(B) Over-Frequency Boosting
(C) Capacitive Power Factor Override
(D) Maximum Power Point Tracking

72. What is the role of Maximum Power Point Tracking (MPPT) algorithms in solar power engineering?

(A) Dynamically adjust inverter input impedance to extract maximum instantaneous power from the non-linear I-V curve of solar modules
(B) Rotate panels mechanically
(C) Cool solar cells
(D) Filter harmonics

73. Which MPPT algorithm iteratively increments or decrements array operating voltage and observes the resulting power change?

(A) Perturb and Observe (P&O)
(B) Incremental Conductance
(C) Constant Voltage
(D) Fractional Open Circuit

74. What is the function of Battery Energy Storage Systems (BESS) in 21st-century power grids?

(A) Provides frequency regulation, peak shaving, renewable time-shifting, and black-start capabilities
(B) Replaces transformers
(C) Eliminates circuit breakers
(D) Generates reactive power exclusively

75. What modern power electronic device converts DC from high-voltage DC links using Voltage Source Converter (VSC) technology?

(A) Modular Multilevel Converter (MMC)
(B) Line-Commutated Thyristor Bridge
(C) Rotary Converter
(D) Mechanical Vibrator

76. What mathematical equation describes the rotor dynamics of a synchronous generator during electromechanical power oscillations?

(A) The Swing Equation: $M \frac{d^2 \delta}{dt^2} = P_m - P_e$
(B) The Bernoulli Equation
(C) The Wave Equation
(D) The Maxwell-Faraday Curl Equation

77. In the Swing Equation, what does $P_m - P_e > 0$ physically cause to happen to the generator rotor?

(A) Net accelerating torque increases rotor speed, causing power angle δ to advance
(B) Rotor decelerates
(C) Rotor reverses direction
(D) Output voltage drops to zero

78. What graphical stability criterion determines whether a synchronous machine will maintain transient stability following a fault clearance?

(A) Equal Area Criterion ($A_{acc} = A_{dec}$)
(B) Nyquist Stability Plot
(C) Routh-Hurwitz Array
(D) Bode Phase Margin

79. What is Critical Clearing Time (CCT) in power system transient stability?

(A) The maximum time duration a fault can persist before being cleared without the system losing synchronism
(B) The time to bill customers
(C) The time to start a turbine from cold
(D) The operating time of a fuse

80. What phenomenon involves dangerous torsional mechanical shaft oscillations triggered by series capacitor compensation interacting with turbine generators?

(A) Subsynchronous Resonance (SSR)
(B) Corona effect
(C) Skin effect
(D) Proximity effect

81. What devices fall under Flexible AC Transmission Systems (FACTS) using power electronics to control power flow dynamically?

(A) STATCOM, SVC, TCSC, and UPFC
(B) Step-down transformers and fuses
(C) Induction motors and alternators
(D) Wattmeters and energymeters

82. What is a STATCOM (Static Synchronous Compensator) functionally equivalent to in power systems?

(A) A solid-state shunt-connected Voltage Source Converter acting as a variable reactive power source/sink with ultra-fast sub-cycle response
(B) A mechanically switched capacitor bank
(C) A series resistor
(D) An overhead ground wire

83. What does Unified Power Flow Controller (UPFC) uniquely achieve in transmission systems?

(A) Simultaneously and independently controls active power, reactive power, and bus voltage on a transmission line
(B) Generates bulk electrical energy
(C) Interrupts fault current in 0 ms
(D) Acts as an infinite bus generator

84. What is the primary cause of Voltage Collapse in heavily loaded power transmission systems?

(A) Inability of the network to meet the escalating reactive power demand, causing monotonic voltage degradation beyond the 'nose point' of P-V curves
(B) Excessive generation of reactive power
(C) Lightning striking the neutral wire
(D) High grid frequency

85. What mathematical curve plots receiving-end bus voltage against active power transfer to identify voltage stability limits?

(A) P-V (Power-Voltage) Nose Curve
(B) I-V Diode Curve
(C) B-H Hysteresis Loop
(D) W?hler S-N Curve

86. In wholesale deregulated electricity markets, what does Locational Marginal Pricing (LMP) reflect?

(A) The marginal cost of supplying the next megawatt-hour of electrical energy at a specific transmission node, incorporating generation, losses, and transmission congestion
(B) A flat nationwide tariff
(C) Average monthly fuel cost
(D) Fixed government subsidy

87. What constitutes 'Ancillary Services' procured by the Independent System Operator (ISO)?

(A) Frequency regulation, spinning reserves, non-spinning reserves, reactive support/voltage control, and black-start capabilities
(B) Janitorial cleaning of substations
(C) Office internet billing
(D) Vehicle fleet maintenance

88. What is 'Black Start' capability in electrical power system restoration?

(A) The ability of a generating unit to start up from a total shutdown without relying on external power from the grid (e.g. diesel/hydro startup)
(B) Operating during a solar eclipse
(C) Running at zero load
(D) Painting switchgear black

89. What standard IEEE reference defines grounding system safety criteria, touch and step potential limits in substations?

(A) IEEE Std 80 (Guide for Safety in AC Substation Grounding)
(B) IEEE Std 519
(C) IEEE Std 1159
(D) IEEE Std 142

90. What is 'Mesh Potential' in substation grounding grid design?

(A) The maximum touch voltage between an earthed grounded metallic structure and the ground surface inside a mesh grid opening during a ground fault
(B) Ground rod resistance
(C) Phase to phase voltage
(D) Neutral displacement

91. What standard IEEE 519-2022 sets limits on in power electrical systems?

(A) Harmonic Voltage and Current Distortion Limits at the Point of Common Coupling (PCC)
(B) Substation noise levels
(C) Tower mechanical tension
(D) Insulator flashover distances

92. How is Total Harmonic Distortion (THD) of a voltage waveform mathematically expressed?

(A) $\text{THD} = \frac{\sqrt{\sum_{n=2}^\infty V_n^2}}{V_1} \times 100\%$
(B) $\text{THD} = \frac{V_1}{\sum V_n}$
(C) $\text{THD} = \frac{V_{max}}{V_{min}}$
(D) $\text{THD} = \frac{P_{active}}{P_{reactive}}$

93. What electrical phenomenon occurs when high-frequency switching transients produce steep voltage wave fronts traveling along motor cables?

(A) Reflected wave voltage doubling ($2 V_{dc}$) at motor terminals due to impedance mismatch, stressing insulation
(B) Zero-crossing failure
(C) DC saturation of stator
(D) Thermal runaway of copper

94. What grid architecture replaces traditional central synchronous generation with inverter-based renewable resources?

(A) Low-Inertia or Zero-Inertia Inverter-Dominated Power Grid
(B) Superheated Steam Cycle
(C) Flywheel-only microgrid
(D) DC battery-isolated system

95. What control technique enables power electronic inverters to mimic the physical inertia of synchronous machines?

(A) Grid-Forming Inverter with Virtual Synchronous Machine (VSM) control
(B) Grid-Following Phase-Locked Loop control only
(C) Fixed duty-cycle PWM
(D) Hysteresis current control

96. What is the primary operational difference between 'Grid-Following' (GFL) and 'Grid-Forming' (GFM) inverters?

(A) GFL tracks an existing grid voltage angle via PLL; GFM establishes its own internal voltage and frequency reference, functioning in 100% inverter grids
(B) GFL generates reactive power only
(C) GFM is only used for DC motors
(D) There is no operational difference

97. What is the purpose of an electromagnetic transient simulation program like PSCAD/EMTDC compared to phasor-based simulators?

(A) Simulates microsecond-level detailed non-linear electromagnetic wave propagation, switching surges, and converter control transients
(B) Calculates yearly utility profit
(C) Designs building air conditioning
(D) Estimates transformer shipping weight

98. What does the 'Smart Grid' conceptual framework incorporate that distinguishes it from the 20th-century legacy grid?

(A) Two-way flows of electricity and real-time digital information, automated self-healing, pervasive sensor telemetry, and dynamic demand response
(B) Higher line voltages only
(C) Bigger coal boilers
(D) Thicker wooden poles

99. What cybersecurity framework standard addresses Critical Infrastructure Protection for bulk electric systems in North America?

(A) NERC CIP (North American Electric Reliability Corporation Critical Infrastructure Protection)
(B) ISO 9001
(C) HIPAA
(D) PCI-DSS

100. What is the overarching ethical responsibility of a licensed Professional Electrical Engineer (PE) according to the IEEE Code of Ethics?

(A) To hold paramount the safety, health, and welfare of the public, and to strive to comply with ethical design and sustainable practices
(B) To maximize corporate shareholder quarterly profit
(C) To patent all discoveries exclusively
(D) To avoid participating in public policy debates

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Detail the historical genesis of electrical engineering from Alessandro Volta's chemical cell to Michael Faraday's law of electromagnetic induction.

Q2. Analyze the historic 'War of the Currents' between Thomas Edison's DC system and George Westinghouse & Nikola Tesla's AC system.

Q3. Explain the mathematical formulation of 3-phase AC power systems, detailing the derivation of line versus phase relationships in Star and Delta configurations.

Q4. Explain the core pillars of Electrical Engineering curricula: Circuits, Machines, Power Systems, Power Electronics, and Control.

Q5. Discuss the Outcome-Based Education (OBE) framework, Washington Accord, Program Outcomes (POs), and Bloom's Taxonomy in technical education.

Q6. Formulate the power-angle relationship of a synchronous transmission line and discuss the physical causes and mitigation of transient instability.

Q7. Explain the physical origin, mathematical modeling, and operational mitigation of the Ferranti Effect in long EHV transmission lines.

Q8. Compare High-Voltage Direct Current (HVDC) transmission with Extra-High-Voltage AC (EHVAC), analyzing technical, economic, and operational break-even points.

Q9. Detail the architecture of modern Substation Automation Systems (SAS) under the IEC 61850 standard, including Process, Bay, and Station bus levels.

Q10. Analyze the electrical power transition challenges: Duck Curve, declining system inertia, and the role of Grid-Forming inverters.

Q11. Explain the principles, construction, and operation of Sulfur Hexafluoride (SF_6) and Vacuum circuit breakers.

Q12. Describe the operating principles and protection zones of Distance Relaying for transmission lines.

Q13. Explain the phenomenon of Corona in EHV overhead lines: Peek's formula, power loss, and reduction methods.

Q14. Detail the working of a Modular Multilevel Converter (MMC) for Voltage Source Converter (VSC) HVDC systems.

Q15. Formulate the symmetrical component transformation and explain its utility in analyzing unbalanced grid faults.

Q16. Explain the design principles of substation grounding grids according to IEEE Std 80.

Q17. Discuss the operating principles of Automatic Generation Control (AGC) and Area Control Error (ACE).

Q18. Detail the principles of Photovoltaic Maximum Power Point Tracking (MPPT) using Incremental Conductance algorithm.

Q19. Explain the electrical architecture of a Utility-Scale Battery Energy Storage System (BESS).

Q20. Describe the IEEE Code of Ethics and its fundamental ethical obligations for practicing electrical engineers.

Q21. Explain the construction and operation of a Synchronous Condenser for grid voltage support and inertia.

Q22. Derive the torque-speed equation for a 3-phase induction motor and explain the maximum torque condition.

Q23. Explain the principles of Wide Area Monitoring Systems (WAMS) using Phasor Measurement Units (PMUs).

Q24. Explain the causes and characteristics of Subsynchronous Resonance (SSR) in series-compensated power grids.

Q25. Explain the operation and characteristics of a Unified Power Flow Controller (UPFC).

Q26. Detail the principles of arc flash hazard analysis according to NFPA 70E and IEEE Std 1584.

Q27. Explain the harmonic distortion limits and Point of Common Coupling (PCC) in IEEE Std 519.

Q28. Describe the primary frequency control mechanisms in conventional thermal and hydro generating units.

Q29. Explain the design and operation of a Double-Fed Induction Generator (DFIG) in wind energy conversion systems.

Q30. Discuss the fundamental principles of engineering design, safety margins, and life-cycle assessment in electrical projects.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [A]
Q2: [A]
Q3: [A]
Q4: [A]
Q5: [A]
Q6: [A]
Q7: [A]
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Q96: [A]
Q97: [A]
Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[illegible]

using polyphase AC. In 1895, the Adams Power Plant at Niagara Falls was commissioned, generating two-phase AC power that was stepped up to 11 kV and transmitted 35 kilometers to Buffalo, New York. This decisive engineering victory established polyphase AC as the global standard for bulk power generation, transmission, and utilization.

[illegible]

vector diagrams, and efficiency calculations.

3. Power Systems & High Voltage Engineering:
Deals with macroscopic generation, transmission, and distribution. Encompasses line inductance/capacitance parameters, GMD/GMR calculations, load flow algorithms (Gauss-Seidel, Newton-Raphson), symmetrical and unsymmetrical fault analysis, protection relaying, and transient stability.

4. Power Electronics & Industrial Drives:
Focuses on solid-state power switches (Thyristors, MOSFETs, IGBTs, SiC/GaN devices) and energy converters: AC-DC controlled rectifiers, DC-DC buck/boost choppers, DC-AC inverters, and PWM control for electric vehicle powertrains.

5. Control Systems & Automation:
Analyzes feedback regulation, transfer functions, state-space representations, root locus, Bode/Nyquist stability criteria, and PID tuning, bridging theoretical electrical dynamics with automated robotic and industrial manufacturing plants.

[illegible]

Power-Angle Relationship and Transient Stability:

Consider a synchronous generator of terminal voltage $E \angle \delta$ connected to an infinite bus $V \angle 0^\circ$ through an equivalent transmission line reactance $X = X_d' + X_{line} + X_{tr}$:
The complex power delivered to the infinite bus is:
$$S = P + j Q = \mathbf{V} \mathbf{I}^* = V \angle 0^\circ \left(\frac{E \angle \delta - V \angle 0^\circ}{j X} \right)^*$$
Expanding the phasor algebra yields the classical Power-Angle equation:
$$P = \frac{E V}{X} \sin\delta = P_{max} \sin\delta$$
where δ is the rotor power angle (rotor displacement relative to the synchronously rotating stator MMF).

Physical Mechanism of Transient Instability:
During a 3-phase fault near the generator bus, terminal voltage collapses ($V \approx 0$), dropping electrical output power $P_e \to 0$. However, mechanical turbine input power P_m cannot change instantaneously. By the Swing Equation ($M \ddot{\delta} = P_m - P_e$), a massive net accelerating torque accelerates the rotor, causing δ to surge.

Equal Area Criterion & Critical Clearing Time (CCT):
Stability requires that the decelerating energy area after fault clearance (A_{dec}) must exceed or equal the accelerating energy area during the fault (A_{acc}). If the protective relay fails to clear the fault before the Critical Clearing Angle δ_{cr}, synchronism is irrevocably lost, tripping the generator.

Mitigation Measures: High-speed numerical relaying (< 2 cycles), fast static excitation systems with Power System Stabilizers (PSS), series capacitor compensation to decrease X, and single-pole auto-reclosing.

>> **Solution to Question 7:**

>> Problem Statement: Explain the physical origin, mathematical modeling, and operational mitigation of the Ferranti Effect in long EHV transmission lines.

3. Physical Mechanism:
The distributed line capacitance draws a continuous capacitive charging current $I_C = j \omega C V$. This current leads the voltage by 90°. When flowing through the line series inductive reactance X_L, it creates an inductive voltage drop $j I_C X_L = j (j \omega C V) (\omega L) = -\omega^2 L C V$, which is 180° out of phase with the inductive drop of lagging current, effectively adding in-phase to the sending voltage.

4. Operational Hazards & Mitigation:
Overvoltage degrades transformer insulation, saturates magnetic cores, and causes dielectric breakdown of switchgear. Mitigation is accomplished by connecting Shunt Reactors (large air-core or gapped-iron inductors) at line terminals and intermediate switching substations to absorb capacitive vars.

[illegible]

1. Process Level:
Comprises physical primary switchgear equipment: power transformers, circuit breakers, disconnectors, and instrument transformers. Non-conventional instrument transformers (NCITs) and Merging Units (MUs) digitize analog CT/PT waveforms into standardized digital Sampled Values (SV, IEC 61850-9-2LE) transmitted at 4000/4800 samples/sec over the Process Bus fiber network.

2. Bay Level:
Houses Intelligent Electronic Devices (IEDs), protection relays, and bay controllers. IEDs receive digital SV streams over the Process Bus to execute differential, overcurrent, and distance protection algorithms. Between IEDs, ultra-high-speed Generic Object Oriented Substation Events (GOOSE) messages execute interlocking and breaker trip commands within sub-4-millisecond latency.

3. Station Level:
Houses the Substation Host HMI, SCADA gateway, engineering workstations, and GPS time-synchronization clocks (IEEE 1588 Precision Time Protocol). The Station Bus (IEC 61850-8-1 MMS protocol) connects Bay IEDs with the station gateway, routing telemetry and control commands to the National Load Dispatch Center.

[illegible]

[Detailed Step-by-Step Answer]:
High-Voltage Circuit Breakers:

1. SF_6 Gas Circuit Breakers:
- Arc Quenching Medium: Sulfur hexafluoride (SF_6) is an electronegative gas possessing exceptional dielectric strength (roughly 2.5 to 3 times that of air at 1 atm) and high thermal conductivity.
- Arc Extinction Process: When contacts part, the intense electric arc decomposes SF_6 into lower fluorides. Due to high electronegativity, SF_6 molecules rapidly capture free electrons, forming heavy, immobile negative ions, removing conductive carriers and building dielectric recovery strength across the contact gap in microseconds.

2. Vacuum Circuit Breakers (VCB):
- Arc Quenching Medium: High vacuum ($< 10^{-6}$ torr). At this pressure, the mean free path of residual gas molecules exceeds contact clearance dimensions.
- Arc Dynamics: The arc consists entirely of ionized metal vapor liberated from the contact surfaces. At current zero, the metal vapor expands and condenses onto surrounding metallic condensation shields within microseconds, restoring dielectric strength almost instantaneously. VCBs are standard for medium-voltage switchgear (11 kV to 33 kV).

>> Solution to Question 12:

[Problem Statement]: Describe the phenomenon of Corona in EHV overhead lines. State Peek's formula for corona loss and reduction methods.

2. Peek's Formula for Critical Disruptive Voltage:
$$V_d = m_0 g_0 \delta r \ln\left(\frac{D}{r} \right)$$
where m_0 is conductor surface irregularity factor, δ is air density correction factor, r is conductor radius, and D is inter-conductor spacing.

3. Corona Reduction Techniques:
- Increasing Conductor Radius: Using hollow conductors or steel-reinforced aluminum conductors.
- Bundled Conductors: Grouping 2, 4, 6, or 8 sub-conductors per phase increases effective radius, drastically reducing electric field gradient at conductor surfaces.

[illegible]

Substation Grounding Grid Design (IEEE Std 80):

1. Safety Objectives:
During a ground fault, large fault currents enter the earth, creating potential gradients along the ground surface. The grounding grid ensures safety limits for:
- Step Potential (E_{step}): Potential difference between a person's feet spaced 1 meter apart without touching any grounded structure.
- Touch Potential (E_{touch}): Potential difference between a person's hand touching a grounded structure and their feet 1 meter away.

2. Tolerable Body Voltage Limits:
$$E_{touch} = \frac{1000 + 1.5 C_s \rho_s}{\sqrt{t_s}} \cdot 0.116$$
Grounding meshes of buried copper conductors and vertical ground rods are engineered to reduce mesh resistance R_g below 1 Ohm and keep touch voltages below human ventricular fibrillation thresholds.

[illegible] algorithm.

[illegible]

>>Solution to Question 2[illegible] grids.

supplies/absorbs real power to the DC link.
2. Series Converter (SSSC): Injects a voltage of controllable magnitude and phase angle in series with the transmission line, independently adjusting transmission line impedance, phase angle, and active/reactive power flow simultaneously.

[illegible] systems.

1. Configuration: Stator is connected directly to the 50/60 Hz grid. Wound rotor is connected to the grid via back-to-back bidirectional IGBT converters (Rotor Side Converter - RSC and Grid Side Converter - GSC) rated for only ~30% of total machine rating.
2. Variable-Speed Operation: Operates in sub-synchronous (speed < synchronous) and super-synchronous (speed > synchronous) regimes, decoupling mechanical rotational speed from grid frequency while providing independent active and reactive power control.

>> Solution to Question 30:

> Problem Statement: Discuss the fundamental principles of engineering design, safety margins, and life-cycle assessment in electrical projects.